Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Artificial Intelligence Application Development Practical

Module - I

Practical – V

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Naïve Bayes' Classification

Using a small dataset (e.g., weather or spam dataset), implement a Naive Bayes classifier using scikit-learn.

DATASET:WEATHER EXAMPLE

Outlook	Temperature	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes

CODE:-

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import GaussianNB
#Dataset
data = {
    'Outlook': ['Sunny', 'Sunny', 'Overcast', 'Rainy', 'Rainy', 'Rainy', 'Overcast'],
    'Temperature': ['Hot', 'Hot', 'Hot', 'Mild', 'Cool', 'Cool', 'Cool'],
    'Humidity': ['High', 'High', 'High', 'High', 'Normal', 'Normal', 'Normal'],
    'Windy': ['False', 'True', 'False', 'False', 'False', 'True', 'True'],
    'Play': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes']}
df = pd.DataFrame(data)
# Label Encoding
le = LabelEncoder()
for col in df.columns:
    df[col] = le.fit_transform(df[col])
# Features and Target
X = df[['Outlook', 'Temperature', 'Humidity', 'Windy']]
y = df['Play']
model = GaussianNB()
model.fit(X, y)
# Test Data
test = pd.DataFrame(
    [[2, 0, 0, 1]],
    columns=['Outlook', 'Temperature', 'Humidity', 'Windy'])
prediction = model.predict(test)
if prediction[0] == 1:
    print("Prediction: Yes, we can play")
else:
    print("Prediction: No, we cannot play")
```

OUTPUT:-

```
Prediction: No, we cannot play
```